

$^{208}\text{Pb}(^{36}\text{S}, ^{35}\text{P}\gamma)$ 2008Wi09,2015Ch56,2019Gr08

2008Wi09,2010WiZZ: A 230-MeV ^{36}S beam was produced by the Argonne Tandem Linac Accelerator System (ATLAS) with an intensity of 1.5 pA on a 0.5 mg/cm² ^{208}Pb target and an intensity of 0.3 pA on a 44 mg/cm² ^{208}Pb target. In the thin-target run, binary transfer products were detected using a heavy-ion parallel-plate avalanche counter (PPAC) array (CHICO) (Time resolution ≈ 0.7 ns). The polar angle covered was 12° – 85° with respect to the beam. γ rays were detected by Gammasphere consisting of 101 HPGe detectors with FWHM=2–10 keV at $E_\gamma=1$ MeV. Event-by-event Doppler shift correction was applied. In the thick-target run, binary transfer products were stopped in the target. γ rays were detected by Gammasphere consisting of 95 HPGe detectors with FWHM=2–3 keV at $E_\gamma=1$ MeV. Measured E_γ , I_γ , $\gamma\gamma$ -coin. Deduced levels. Comparisons with shell-model calculations. Branching-ratio limits were reported for predicted transitions to the $2\hbar\omega$ bandheads in ^{35}P and ^{34}Si . An e-mail reply from Mathis Wiedeking in April, 2010 (**2010WiZZ**) provides relative γ -ray intensities, supplementing **2008Wi09**.

2015Ch56: A 215-MeV ^{36}S beam was produced using the combination of XTU tandem Van de Graaff accelerator and ALPI superconducting linear accelerator at the INFN Legnaro National Laboratory. The target was 300- $\mu\text{g}/\text{cm}^2$ 99.7% enriched ^{208}Pb on a 20 $\mu\text{g}/\text{cm}^2$ carbon backing. Projectile-like fragments produced in multinucleon binary grazing reactions were separated and identified by the PRISMA spectrometer. γ rays were detected using the CLARA array of 22 EUROBALL escape-suppressed HPGe clover detectors. Doppler corrections of γ -ray energies were performed event by event. Measured E_γ , I_γ , $(^{35}\text{P})\gamma$ -coin, and $\gamma\gamma$ -coin. **2015Ch56** also revisited the $\gamma\gamma\gamma$ -coin of $^{36}\text{S}+^{176}\text{Yb}$ deep-inelastic data by J. Ollier Ph.D. thesis, University of Paisley (2004) to strengthen the evidence for γ -ray placements (see Ref. [39] in **2015Ch56**). Deduced levels, J , π . Comparisons with shell-model calculations.

2019Gr08: A 225-MeV ^{36}S beam was provided by Tandem-ALPI accelerator complex at the INFN Legnaro National Laboratory. The target was 1 mg/cm² 99.7% enriched ^{208}Pb with 1 mg/cm² Nb backing and mounted onto the Cologne differential plunger. Projectile-like fragments produced in binary grazing reactions were separated and identified by the PRISMA spectrometer. γ rays were detected using the AGATA demonstrator array of five triple cluster modules of 36-fold segmented Ge crystals covering backward angles from 135° to 175° . Doppler corrections of γ -ray energies were performed event by event. Measured E_γ , $(^{35}\text{P})\gamma$ -coin, and level lifetimes using the differential recoil-distance method (DRDM). Comparison with shell-model calculations.

 ^{35}P Levels

$E(\text{level})^\dagger$	$J^\pi^\ddagger$	$T_{1/2}^\#$	Comments
0	$1/2^+$		
2386.7 7	$3/2^+$	<0.69 ps	$T_{1/2}$: estimated mean lifetime $\tau < 1$ ps (2019Gr08).
3860.8 7	$5/2^+$	<0.69 ps	$T_{1/2}$: estimated mean lifetime $\tau < 1$ ps (2019Gr08).
4102.1 7	$(7/2^-)$	>69 ps	J^π : $7/2^-$ proposed by 2019Gr08 based on comparisons with shell-model calculations. $T_{1/2}$: estimated mean lifetime $\tau > 100$ ps (2019Gr08).
4381.9 10	$(5/2^-)$		
4494.2 8	$(7/2^-)$	2.29 ps 49	J^π : $7/2^-$ proposed by 2019Gr08 based on comparisons with shell-model calculations. $T_{1/2}$: measured mean lifetime $\tau = 3.3$ ps 7 (2019Gr08).
4767.1 10	$(9/2^-)$		J^π : $9/2^-$ proposed by 2019Gr08 based on comparisons with shell-model calculations.
4869.0 8	$(5/2^-, 7/2^-)$		
4962.1 12	$(9/2^-)$		J^π : $9/2^-$ proposed by 2019Gr08 based on comparisons with shell-model calculations.
5089.8 11	$(11/2^-)$		J^π : $11/2^-$ proposed by 2019Gr08 based on comparisons with shell-model calculations.
5488.2 10			
5560.1 12	$(5/2^-)$		
6222.4 11	$(7/2^-, 9/2^-, 11/2^-)$		

[†] From a least-squares fit to γ -ray energies.

[‡] From the Adopted Levels.

[#] From differential recoil-distance method (DRDM) (**2019Gr08**).

$^{208}\text{Pb}(^{36}\text{S}, ^{35}\text{P}\gamma)$ **2008Wi09,2015Ch56,2019Gr08 (continued)** $\gamma(^{35}\text{P})$

E_γ	I_γ	$E_i(\text{level})$	J_i^π	E_f	J_f^π	Comments
128 1	10 5	5089.8	(11/2 ⁻)	4962.1	(9/2 ⁻)	E_γ : weighted average of 128 1 (2008Wi09) and 127 1 (2015Ch56). I_γ : unweighted average of 14 2 (2008Wi09) and 5.2 6 (2015Ch56).
241 1	61 4	4102.1	(7/2 ⁻)	3860.8	5/2 ⁺	E_γ : From 2008Wi09 and 2015Ch56.
273 1	12.8 8	4767.1	(9/2 ⁻)	4494.2	(7/2 ⁻)	I_γ : From 2008Wi09. Other: 32.6 9 (2015Ch56). E_γ : From 2008Wi09 and 2015Ch56.
322 1	20 7	5089.8	(11/2 ⁻)	4767.1	(9/2 ⁻)	I_γ : weighted average of 12 2 (2008Wi09) and 12.9 8 (2015Ch56). E_γ : weighted average of 321 1 (2008Wi09) and 323 1 (2015Ch56). I_γ : unweighted average of 27 3 (2008Wi09) and 12.9 8 (2015Ch56).
374 [†] 1	3 [†] 1	4869.0	(5/2 ⁻ , 7/2 ⁻)	4494.2	(7/2 ⁻)	E_γ : weighted average of 391 1 (2008Wi09) and 392 1 (2015Ch56). I_γ : unweighted average of 35 3 (2008Wi09) and 24.9 11 (2015Ch56).
392 1	30 5	4494.2	(7/2 ⁻)	4102.1	(7/2 ⁻)	
468 2	16.2 12	4962.1	(9/2 ⁻)	4494.2	(7/2 ⁻)	E_γ : unweighted average of 466 1 (2008Wi09) and 469 1 (2015Ch56). I_γ : weighted average of 14 2 (2008Wi09) and 16.8 11 (2015Ch56).
487 [†] 1	<2 [†]	4869.0	(5/2 ⁻ , 7/2 ⁻)	4381.9	(5/2 ⁻)	I_γ : 1 1 from 2008Wi09. E_γ : weighted average of 632 1 (2008Wi09) and 633 1 (2015Ch56). I_γ : unweighted average of 6 1 (2008Wi09) and 10.4 9 (2015Ch56).
633 1	8.2 22	4494.2	(7/2 ⁻)	3860.8	5/2 ⁺	
664 1	32 15	4767.1	(9/2 ⁻)	4102.1	(7/2 ⁻)	E_γ : weighted average of 663 1 (2008Wi09) and 665 1 (2015Ch56). I_γ : unweighted average of 47 4 (2008Wi09) and 17.8 10 (2015Ch56).
767 [†] 1	5 [†] 1	4869.0	(5/2 ⁻ , 7/2 ⁻)	4102.1	(7/2 ⁻)	E_γ : unweighted average of 856 1 (2008Wi09) and 861 1 (2015Ch56). I_γ : weighted average of 13 2 (2008Wi09) and 9.8 12 (2015Ch56).
859 3	10.7 14	4962.1	(9/2 ⁻)	4102.1	(7/2 ⁻)	
993 [†] 1	5 [†] 1	5488.2		4494.2	(7/2 ⁻)	E_γ, I_γ : From 2015Ch56.
1009 [†] 1	<1 [†]	4869.0	(5/2 ⁻ , 7/2 ⁻)	3860.8	5/2 ⁺	
1132 [†] 1	<1 [†]	6222.4	(7/2 ⁻ , 9/2, 11/2 ⁻)	5089.8	(11/2 ⁻)	E_γ, I_γ : From 2015Ch56.
1260 [†] 1	4 [†] 1	6222.4	(7/2 ⁻ , 9/2, 11/2 ⁻)	4962.1	(9/2 ⁻)	
^x 1353 1	9.2 11					E_γ, I_γ : From 2015Ch56.
1387 [†] 1	3 [†] 1	5488.2		4102.1	(7/2 ⁻)	
1458 [†] 1	7 [†] 2	5560.1	(5/2 ⁻)	4102.1	(7/2 ⁻)	E_γ : weighted average of 1473 1 (2008Wi09) and 1474 1 (2015Ch56). I_γ : weighted average of 15 2 (2008Wi09) and 16.4 14 (2015Ch56).
1474 1	15.9 14	3860.8	5/2 ⁺	2386.7	3/2 ⁺	
^x 1592 1	7.7 10					E_γ, I_γ : From 2015Ch56.
1715 [†] 1	4 [†] 1	4102.1	(7/2 ⁻)	2386.7	3/2 ⁺	E_γ : weighted average of 1995 1 (2008Wi09) and 1994 1 (2015Ch56, unplaced).
1729 [†] 1	4 [†] 1	6222.4	(7/2 ⁻ , 9/2, 11/2 ⁻)	4494.2	(7/2 ⁻)	
1995 1	8 6	4381.9	(5/2 ⁻)	2386.7	3/2 ⁺	

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$^{208}\text{Pb}(^{36}\text{S}, ^{35}\text{P}\gamma)$ [2008Wi09](#), [2015Ch56](#), [2019Gr08](#) (continued) $\gamma(^{35}\text{P})$ (continued)

E_γ	I_γ	$E_i(\text{level})$	J_i^π	E_f	J_f^π	Comments
2386 <i>I</i>	30 <i>4</i>	2386.7	$3/2^+$	0	$1/2^+$	I_γ : unweighted average of 2 <i>I</i> (2008Wi09) and 14.2 <i>II</i> (2015Ch56). E_γ : From 2008Wi09 and 2015Ch56 . I_γ : From 2008Wi09 . Other: 99.2 <i>28</i> (2015Ch56). Shell-model calculations indicate a small occupancy of the proton $1d_{3/2}$ orbit in the ground state of ^{36}S .
3861 <i>I</i>	100.0 <i>32</i>	3860.8	$5/2^+$	0	$1/2^+$	E_γ : weighted average of 3861 <i>I</i> (2008Wi09) and 3860 <i>I</i> (2015Ch56). I_γ : From 2015Ch56 . Other: 100 (2008Wi09).
4102 [†] <i>I</i>	33 [†] <i>5</i>	4102.1	$(7/2^-)$	0	$1/2^+$	E_γ : other: 4101 (2015Ch56).

[†] From [2008Wi09](#).^x γ ray not placed in level scheme.

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Level Scheme

Intensities: Relative I_γ

Legend

- $\longrightarrow$ $I_\gamma < 2\% \times I_\gamma^{\text{max}}$
 $\longrightarrow$ $I_\gamma < 10\% \times I_\gamma^{\text{max}}$
 $\longrightarrow$ $I_\gamma > 10\% \times I_\gamma^{\text{max}}$

